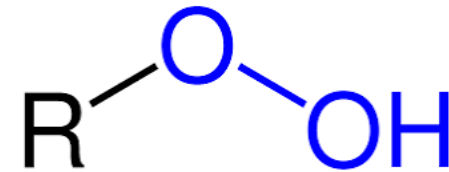


Simulation of Solar Photolysis of Organic Hydroperoxides ROOH (tert-butyl hydroperoxide)

Helen Clifford
CyberTraining Workshop 2026

Background and Motivation



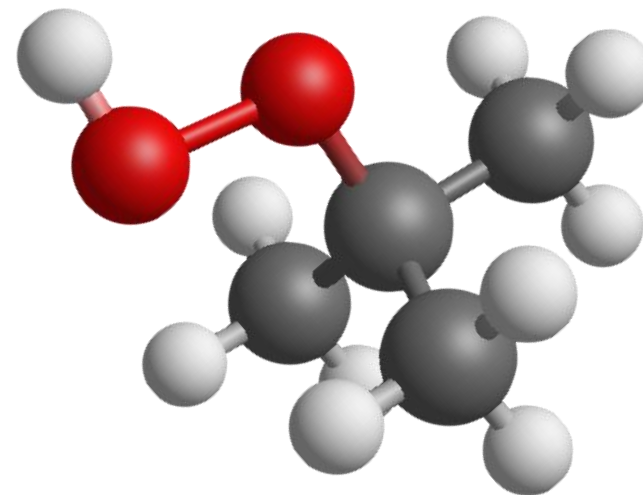
- Organic hydroperoxides serve as an important intermediate in the formation of secondary organic aerosols
- From UV radiation, the O-O bond can dissociate on a repulsive excited state leading to RO+OH radical products
- Experimental evidence shows two possible mechanistic pathways and electronic states for RO-OH dissociation*:
 - Direct dissociation on S_1 suggested to be the dominant pathway but also evidence of internal conversion through S_1/S_0 degeneracy followed by ground-state dissociation*

*Zou, M. et al., J. Phys. Chem. A 2025, 129 (13), 3052–3062

Outline and Research Objective

- Performing NAMD simulations with PySPAWN on tert-butyl hydroperoxide
- **Objective:** After excitation, which electronic surface and through which pathway does tert-butyl hydroperoxide undergo RO–OH bond homolysis?
 - O–O homolysis occurs on S_1 or after transition through S_1/S_0 internal conversion

System of Interest:

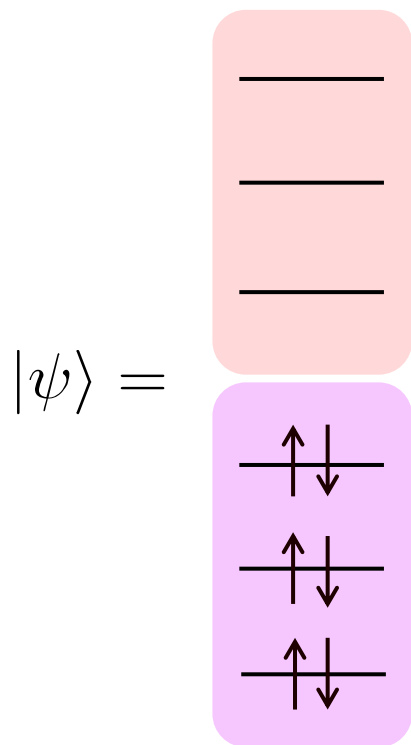


tert-butyl
hydroperoxide
(TBHP)

Electronic Structure Methods

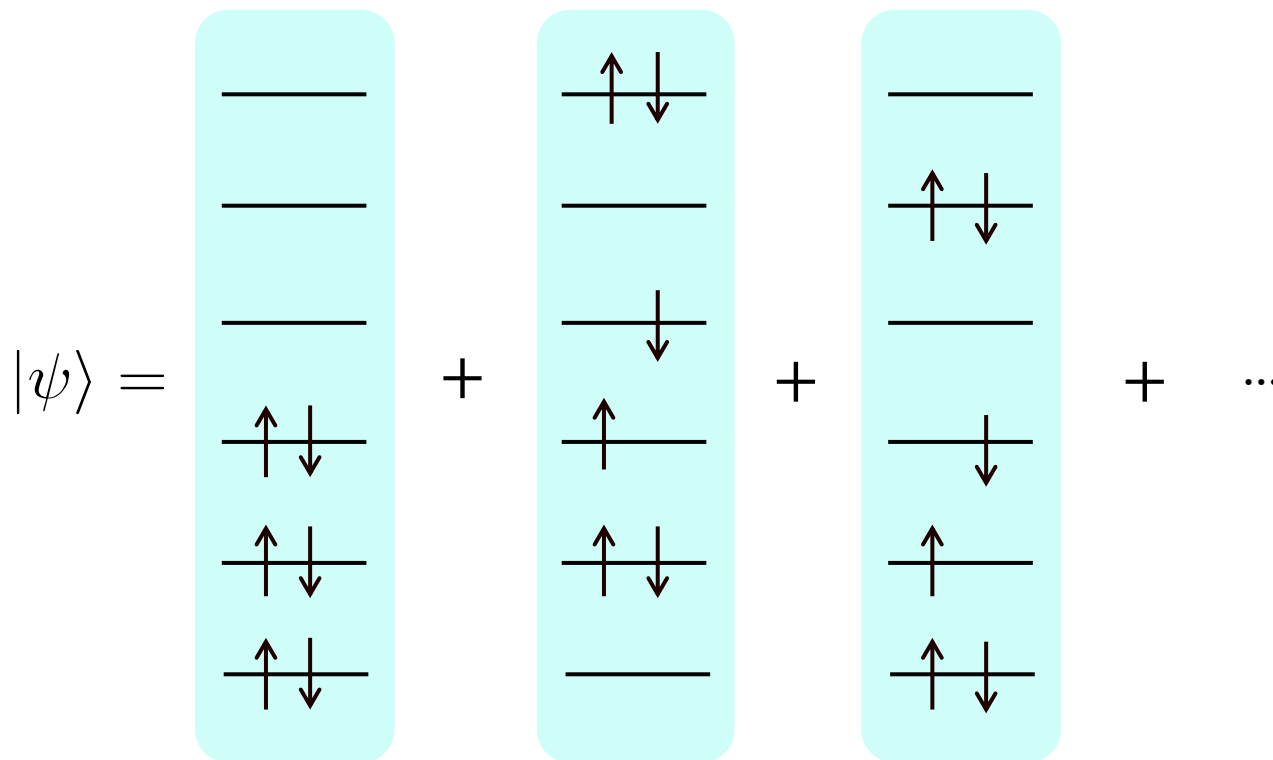
Single Reference:

HF, KS-DFT



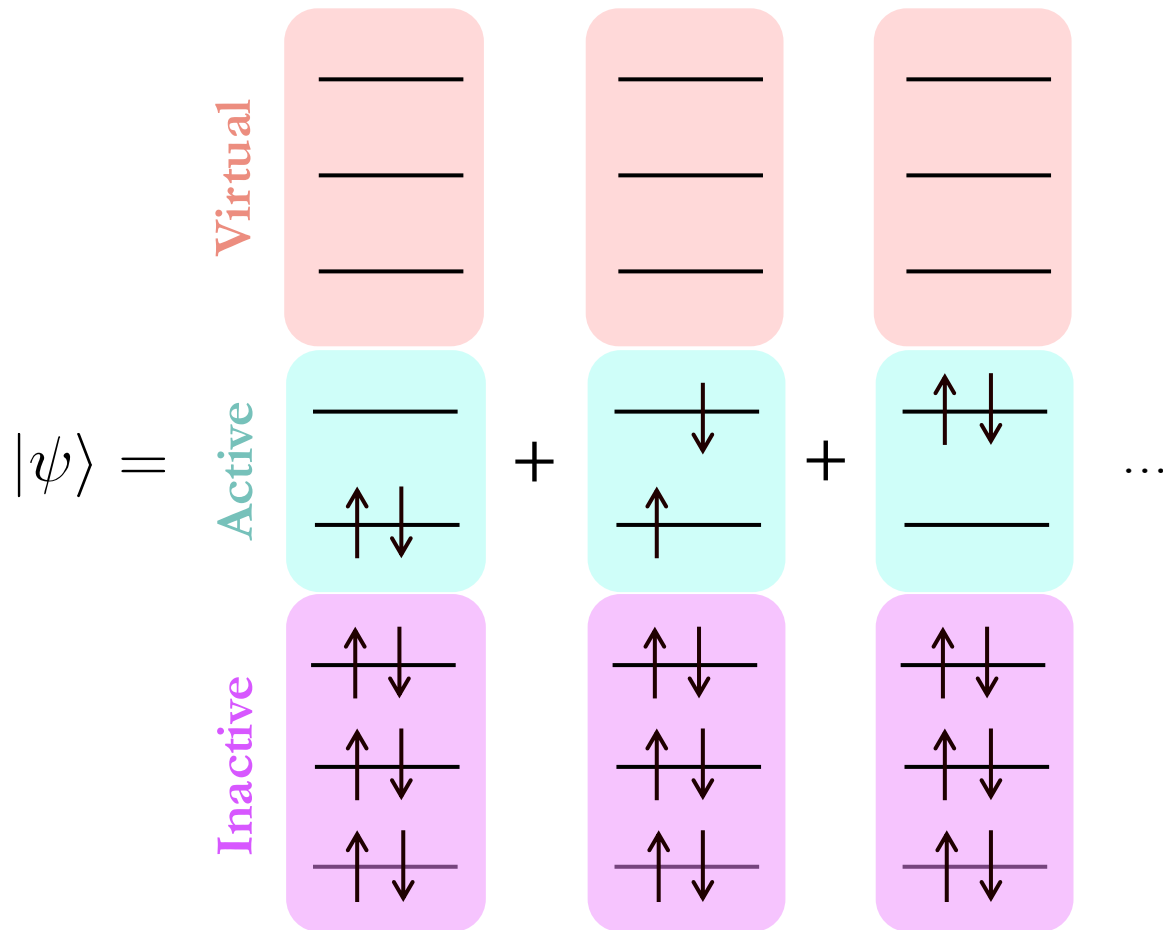
Multi-Reference:

FCI, MCSCF, MRCI



Complete Active Space Self Consistent Field (CASSCF)

Complete Active Space Wave Function



CASSCF:

- Recovers correlation internal but misses correlation external to active space
- Nontrivial choice of active space

Post CASSCF:

- Aim to recover remaining external correlation

Ab Initio Multiple Spawning (AIMS)

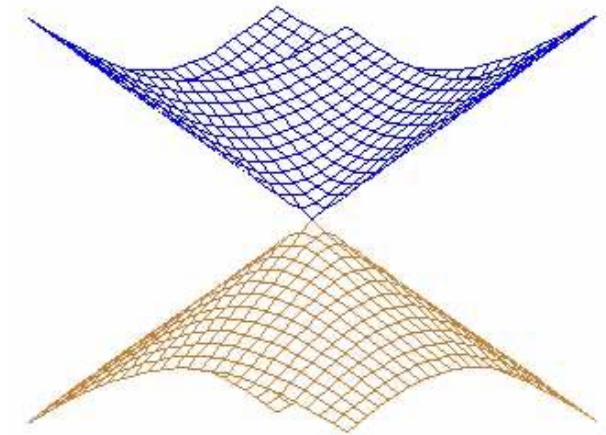
- AIMS represents the wave function as a sum over states of the product of the time-dependent nuclear wave function $\chi_I(\mathbf{R}, t)$ and time-independent electronic wave function $\phi_I(r, \mathbf{R})$

$$\psi = \sum_I \chi_I(\mathbf{R}, t) \phi_I(r, \mathbf{R})$$

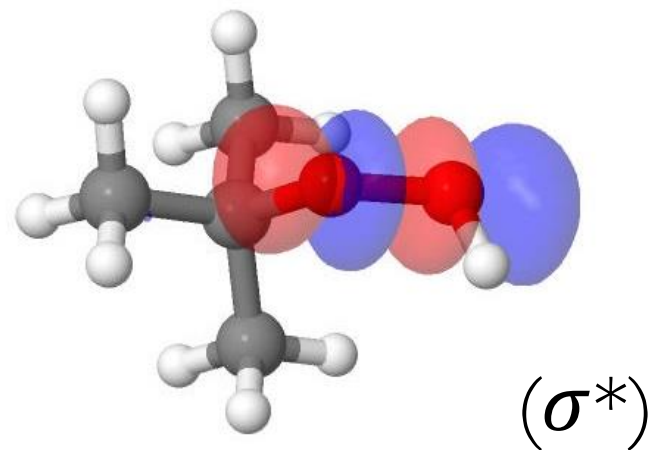
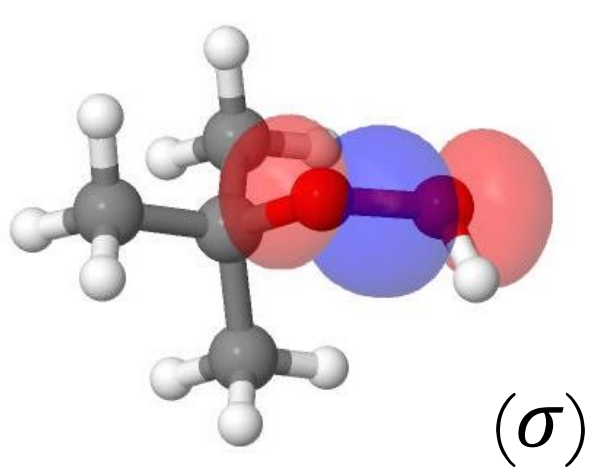
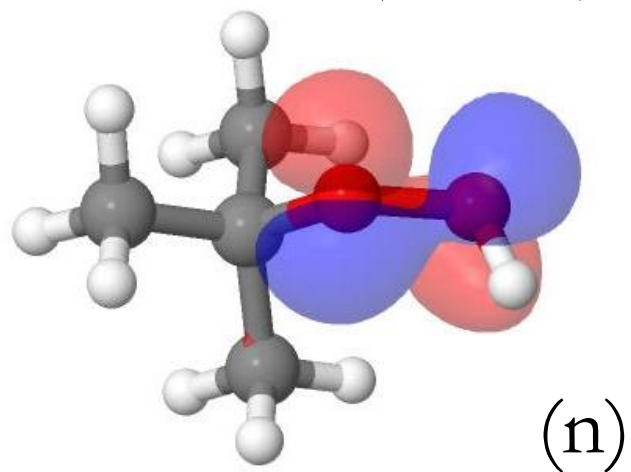
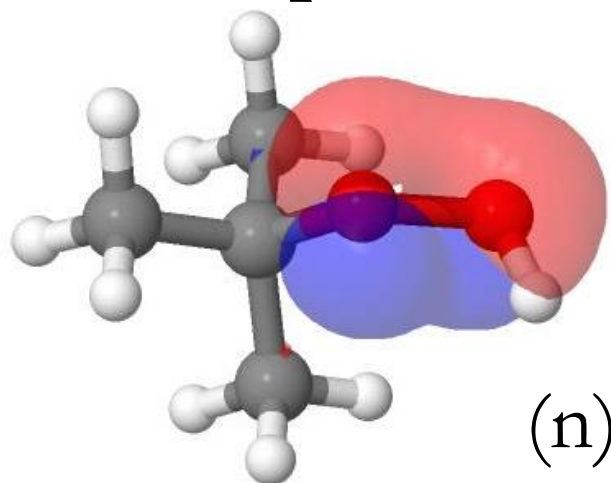
- The nuclear wave function is a linear combo of frozen Gaussian basis functions (trajectory basis function = $\chi_i^I(\mathbf{R}; \bar{\mathbf{R}}_i^I(t), \bar{\mathbf{P}}_i^I(t), \alpha_i^I)$)

$$\chi_I(\mathbf{R}, t) = \sum_i^{N_I(t)} c_i^I(t) \chi_i^I(\mathbf{R}; \bar{\mathbf{R}}_i^I(t), \bar{\mathbf{P}}_i^I(t), \alpha_i^I)$$

- When a trajectory basis function is in a region of strong coupling a new trajectory basis function is spawned onto the coupled electronic state

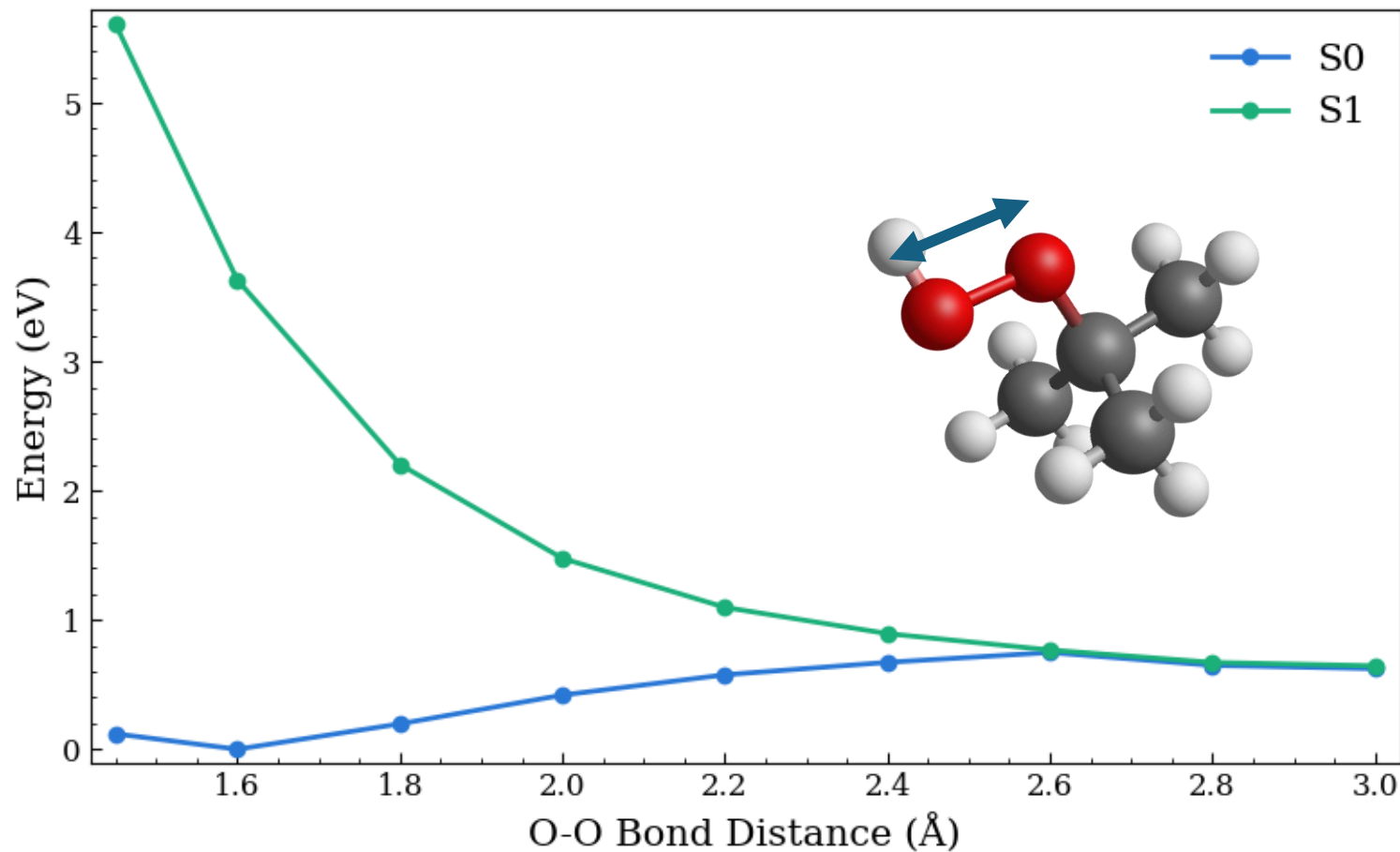


Active Space: SA(4)-CASSCF(6e,4o)



4 orbitals directly involved in the $n \rightarrow \sigma^*$ transition that drives the $S1 \leftarrow S0$ excitation and the O–O homolysis¹

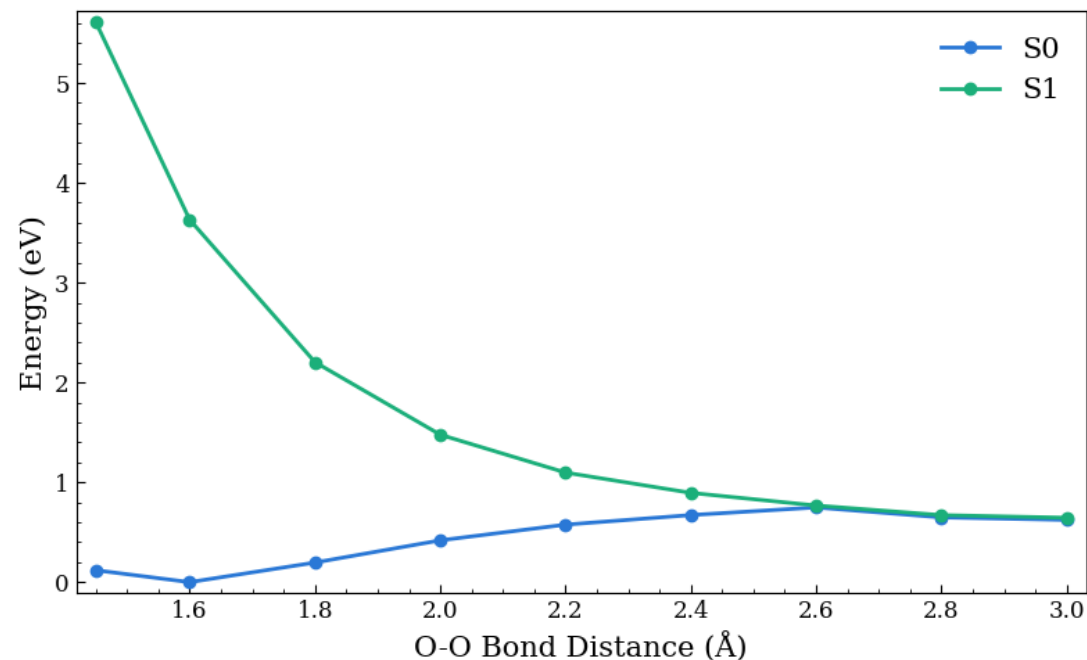
Potential Energy Surface



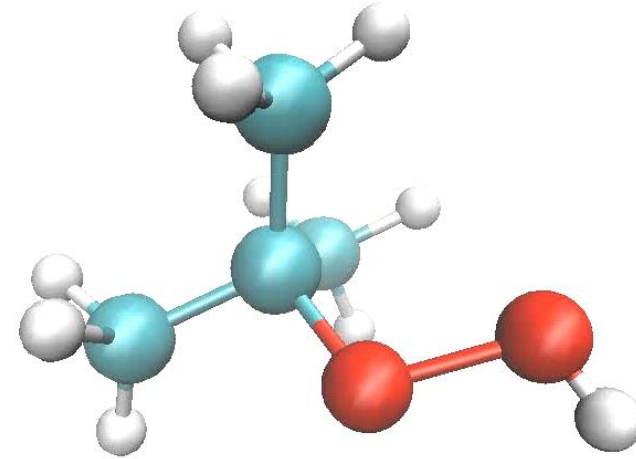
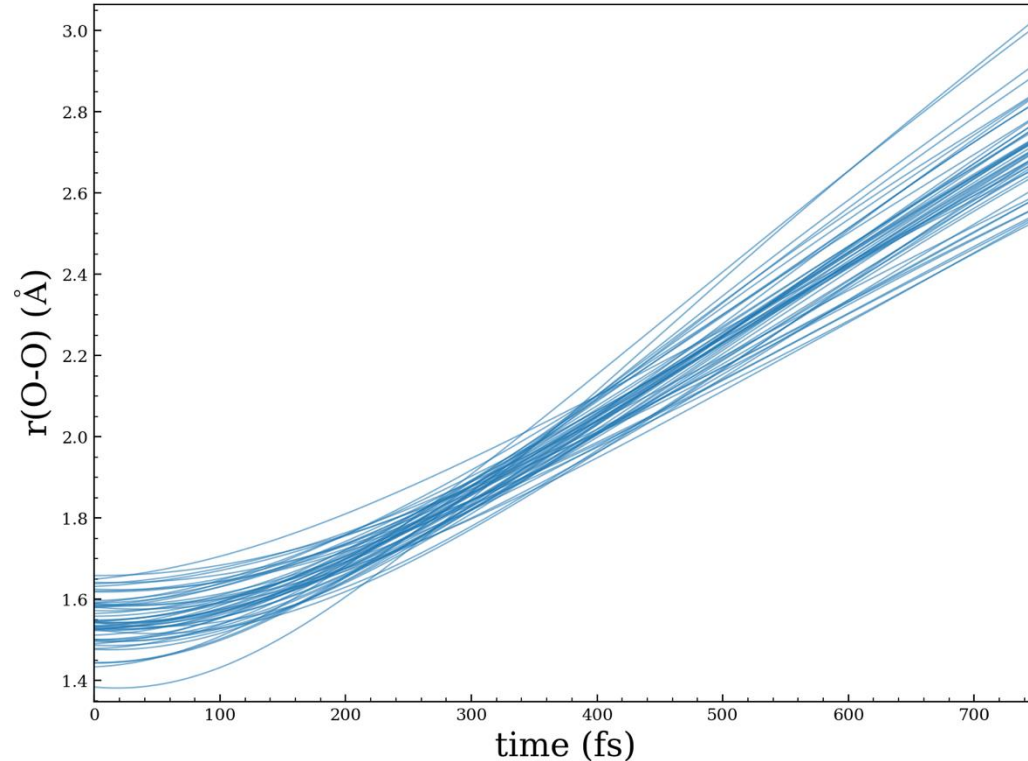
- S₁ is steep and repulsive indicative of rapid dissociation
- S₁ becomes degenerate with S₀ at large O-O distances
- Relaxed curves (with respect to S₁) were computed at SA(4)-CASSCF(6e,4o) level of theory
 - SA(4) since S₂ and S₃ lie close in energy to S₀ and S₁
 - Inclusion of more states helps stabilize and balance along the bond-breaking coordinate

Dynamics Set Up and Details

- Hessian/frequency calculation on S_0 equilibrium geometry for preparing initial conditions from Wigner distribution
- Generated initial conditions through random seed that sets Wigner-sampled momenta and positions
- Propagate SA(4)-CASSCF AIMS dynamics initialized on S_1 (ts = 0.5 fs, timescale = 750 fs)
- Analysis for trajectories
 - Defined a RO-OH bond cleavage threshold of 2.6 Å: operational choice of ~ 1 Å above equilibrium geometry RO-OH bond length (1.4-1.6 Å)
 - 2.6 Å is beyond the steep portion of repulsive S_1 curve and beginning of state degeneracy of S_1/S_0

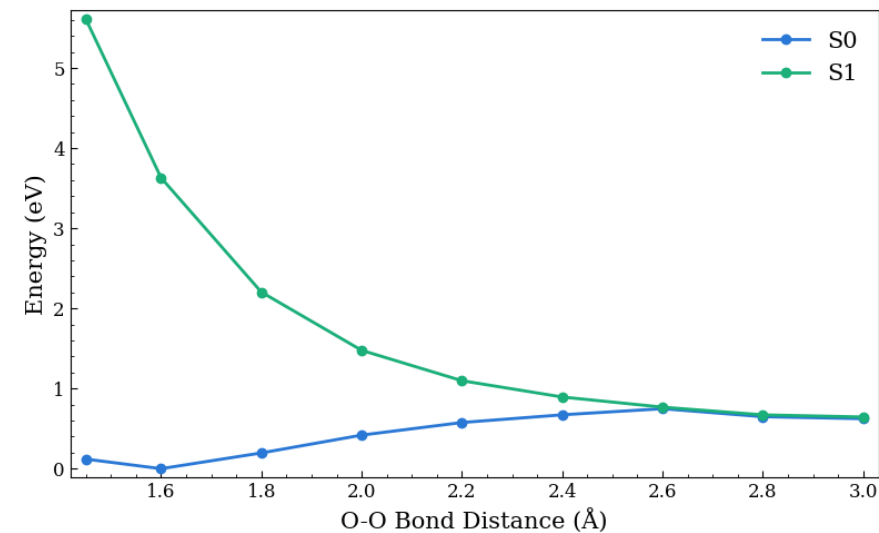
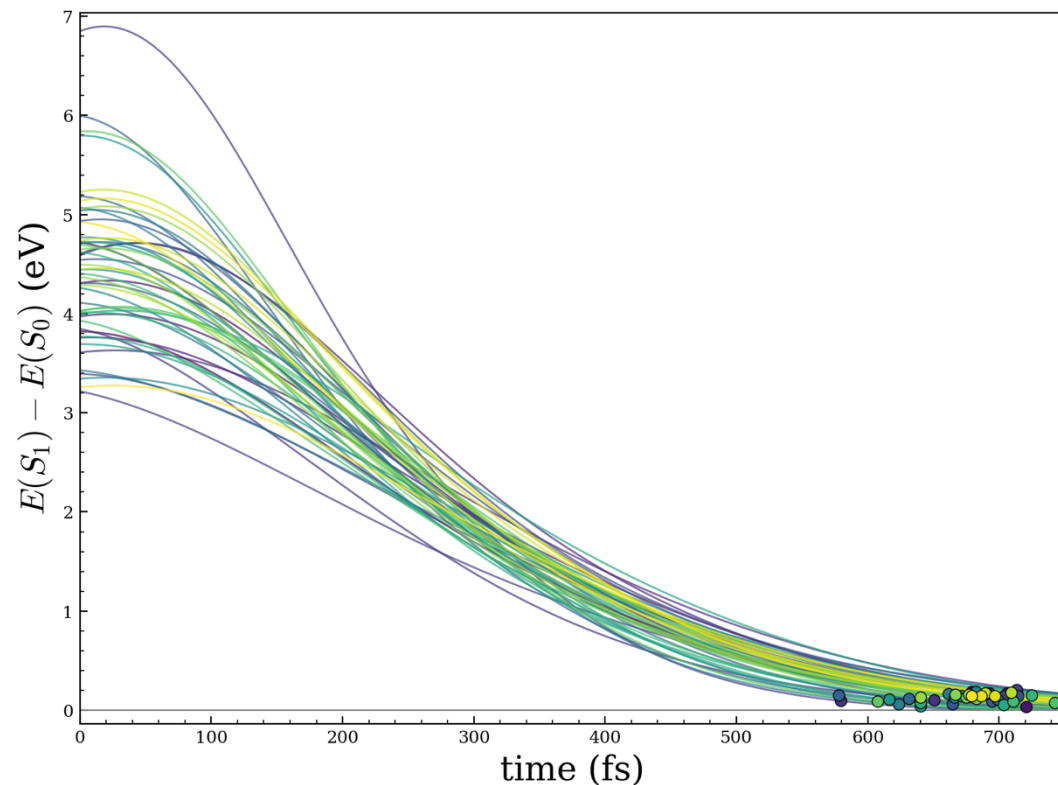


Time evolution of RO-OH bond distance



- All 50 trajectories have continuous RO–OH bond elongation from FC region (1.4–1.6 Å) reaching 2.5–3.0 Å by 750 fs
 - Indicative of the dissociation process on a repulsive S_1 surface

Dynamics Results



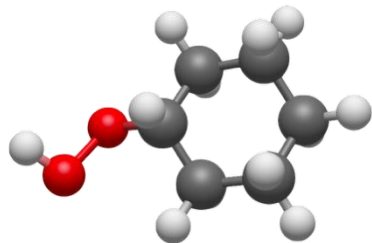
- Time when each trajectory crosses $r(\text{O-O}) = 2.6 \text{ \AA}$ is marked with a dot
- ΔE is large initially and decreases as the RO-OH bond stretches
- Reflects behavior seen in the PES

Interpretation and Results

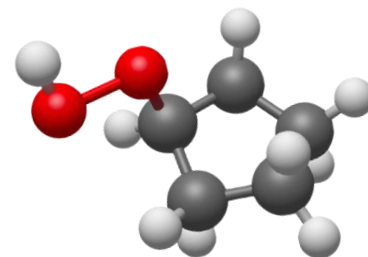
- All 50 trajectories initiated on S_1 remained on S_1 during the 750 fs simulation, no observed spawns to S_0
- Could indicate that RO-OH bond dissociation occurs on S_1 curve rather than through internal conversion and strong coupling to the S_0 along O-O stretching coordinate
- For AIMS, spawning is dependent on the nonadiabatic coupling vector rather than just the electronic state energy gap
 - Near degeneracy shown in the PES may not guarantee spawning onto different electronic states.

Conclusions and Future Work

- AIMS dynamics with SA-CASSCF with 50 trajectories show direct dissociation on S1 surface as the dominant mechanistic pathway along O-O stretching coordinate.
- Current work could indicate predominant sampling from O-O stretching coordinates.
 - Extended sampling from torsional or O-O-H bending coordinates could provide different results
- Additional organic hydroperoxides: cyclohexyl and cyclopentyl
- Running dynamics with more initial conditions
- Higher level of theory for dynamics: CASPT2, PDFT, etc.



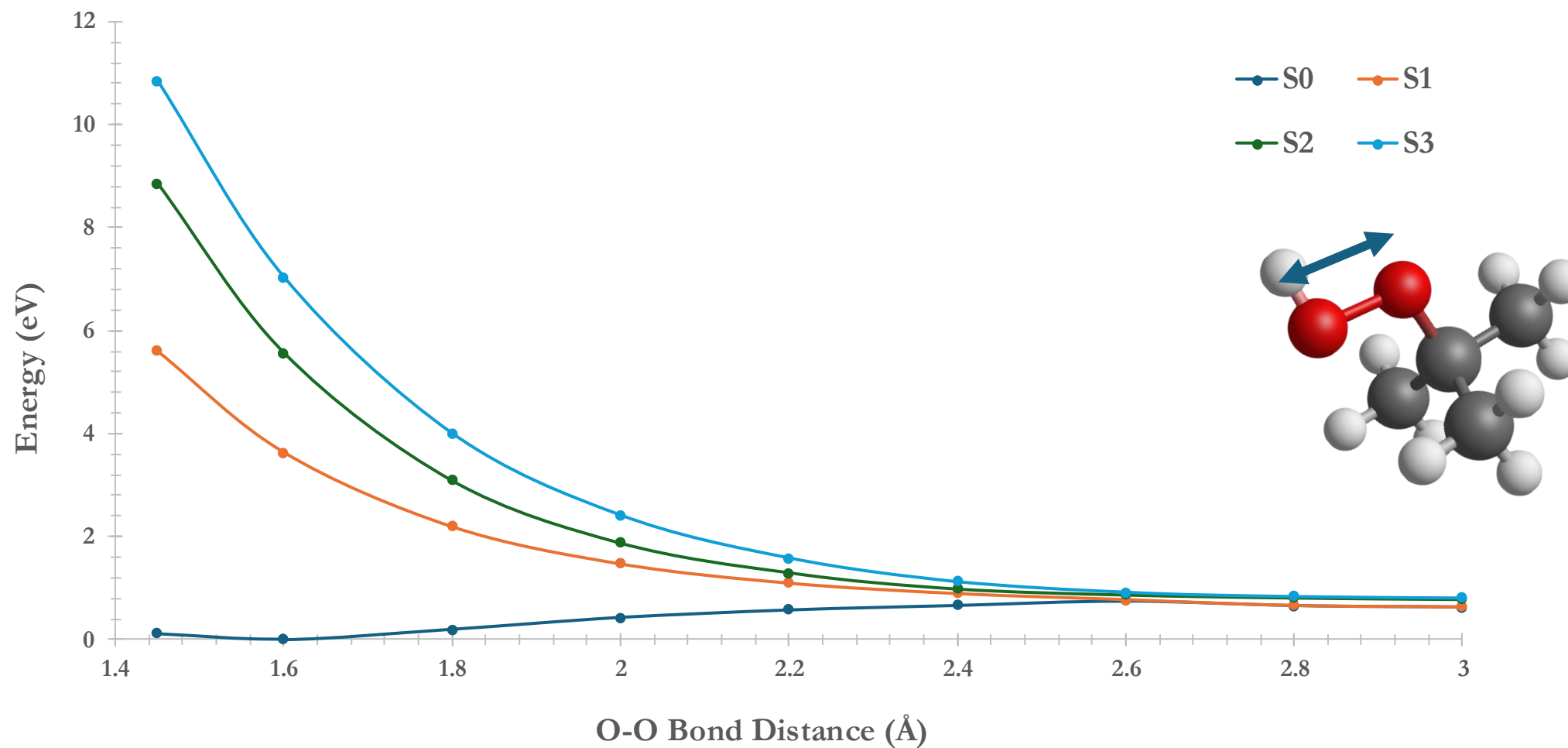
cyclohexyl hydroperoxide (C6H11OOH)



cyclopentyl hydroperoxide (C5H9OOH)

Backup Slides

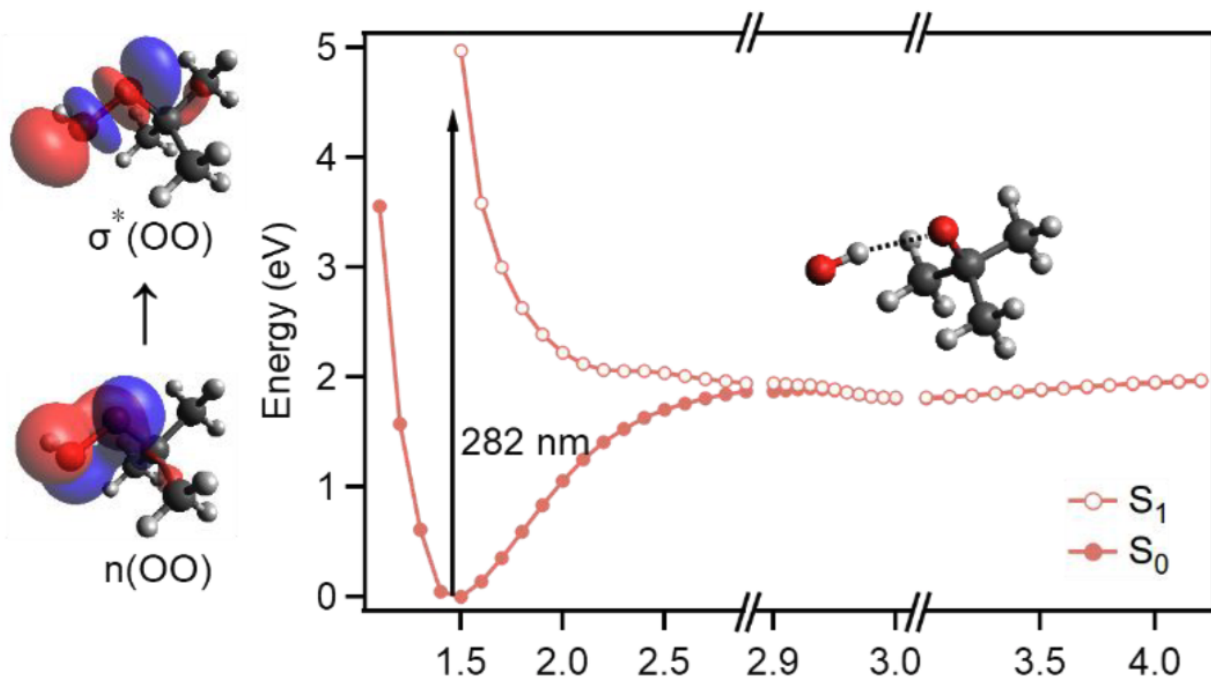
Potential Energy Surface – SA(4)CASSCF



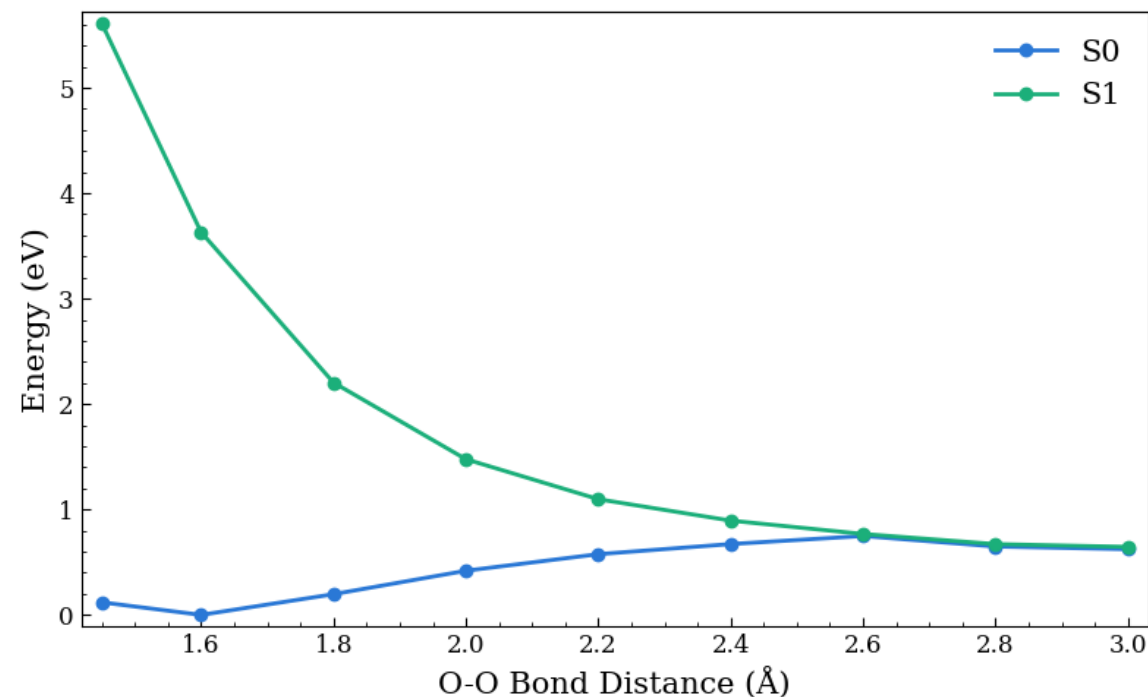
Zero of energy is taken relative to the S_0 minimum

Literature Results vs Current Work

SA(4)- CASPT2(6e,4o)/AVTZ level of theory



SA(4)-CASSCF(6e,4o)/cc-pVDZ level of theory



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